

RetailHub Sales & Performance Analysis

Project Overview

This project demonstrates the use of SQL to analyze retail sales and business performance using a relational database built in MySQL. The database consists of five related tables: Stores, Employees, Customers, Products, and Orders.

The project was designed to simulate real-world business scenarios by answering management questions related to sales performance, customer behavior, employee productivity, and product performance. Various SQL techniques were applied, including joins, aggregate functions, subqueries, CASE statements, views, and date functions. The objective of this project was to transform raw transactional data into meaningful business insights that support decision-making.

Database Structure

The RetailHub database consists of five related tables:

- **Stores** – Stores and their locations.
- **Employees** – Employee details linked to each store.
- **Customers** – Customer information.
- **Products** – Product details, categories, and prices.
- **Orders** – Sales transactions connecting customers, products, employees, and stores.

Relationships between the tables were established using Primary Keys and Foreign Keys to ensure data integrity.

SQL Skills Demonstrated

Throughout this project, the following SQL concepts were applied:

- Database creation
- Table creation
- Primary Keys
- Foreign Keys
- INSERT statements
- INNER JOIN
- LEFT JOIN
- Aggregate Functions (SUM, AVG, COUNT)
- GROUP BY
- HAVING
- ORDER BY
- LIMIT
- Subqueries
- CASE Statements
- Views
- Date Functions (MONTH)

Business Questions Answered

The following business questions were answered using SQL:

- Which store generated the highest revenue?
- Which employee generated the highest revenue?
- Which customer spent the most money?
- Which product category generated the highest revenue?
- Which city generated the highest revenue?

- Which employee processed the highest number of orders?
- What is the average order value for each store?
- Which products are priced above the average product price?
- Which customers spent above the average customer spending?
- How can products and stores be classified using CASE statements?
- What is the total revenue generated each month?
- Which products have never been sold?
- Which customers purchased Electronics products?

Key Findings

The analysis produced several valuable business insights:

- RetailHub Sandton generated the highest overall revenue.
- Ishmael Lehoko was the highest-performing employee based on revenue generated.
- Lerato Lehoko was the highest-spending customer, with a total spending of R32, 100.
- The Networking category generated the highest revenue among all product categories.
- David Johnson and Kagiso Ndlovu processed the highest number of customer orders.
- Some stores recorded higher average order values than others, indicating stronger sales performance.
- Monthly revenue analysis showed how sales were distributed over time.

Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Introduce employee recognition programs, such as performance bonuses or Employee of the Month awards to reward top-performing employees.

2. Increase marketing efforts in lower-performing stores and cities to improve customer traffic and overall sales.
3. Continue investing in high-performing product categories, particularly networking products, while reviewing strategies to improve the sales of lower-performing categories.
4. Monitor average order values across stores to identify opportunities for improving sales performance through product promotions and upselling strategies.

Conclusion

This project demonstrates how SQL can be used to transform raw transactional data into meaningful business insights. By applying database design principles, SQL queries, joins, aggregate functions, subqueries, CASE statements and views, valuable information was extracted to support business decision-making. The project strengthened practical SQL skills while demonstrating the ability to solve real-world business problems using data analysis techniques.